Lab 1 – Intro to Azure

Provided by Microsoft Azure at Reston, VA

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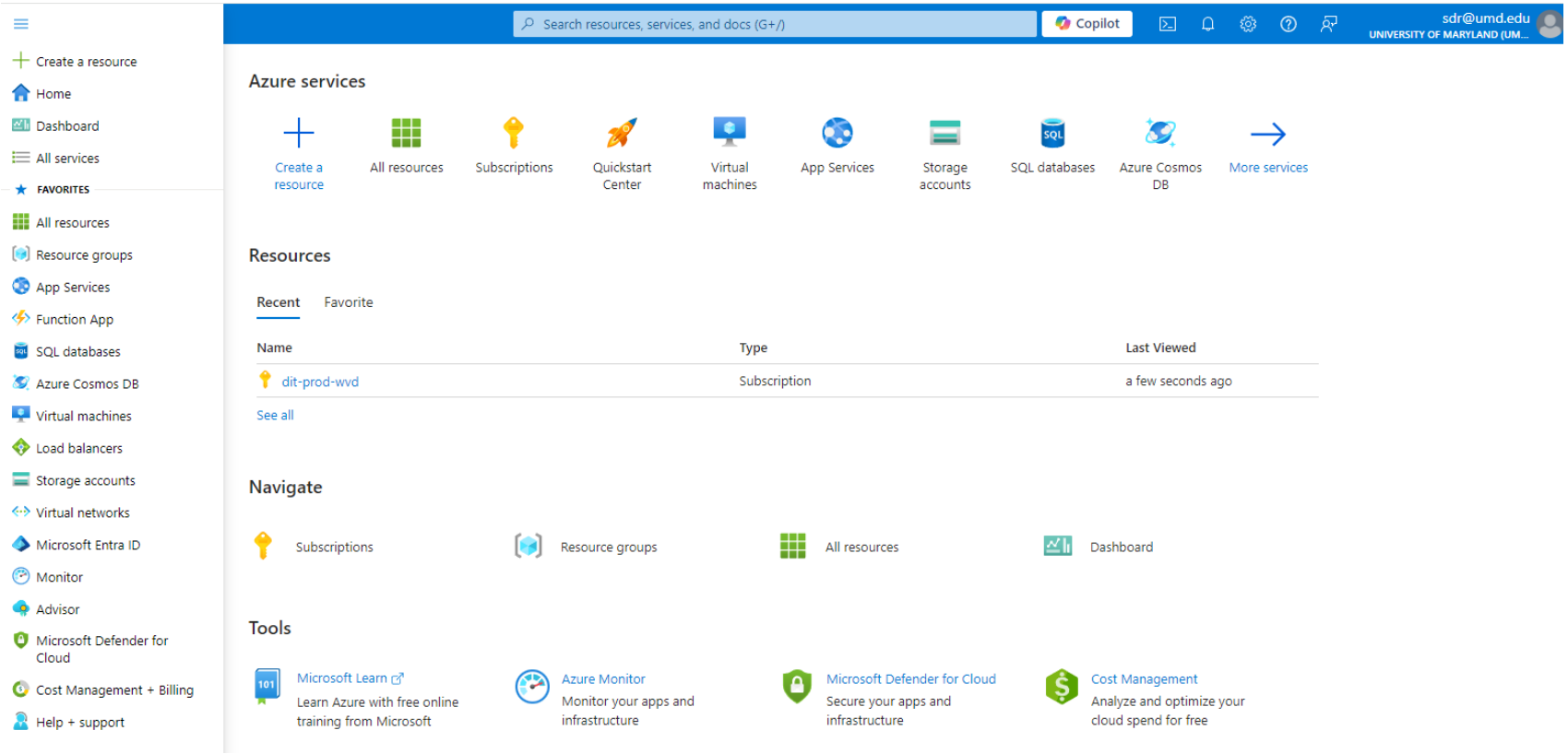
DEPARTMENT OF COMPUTER SCIENCE

Azure Portal

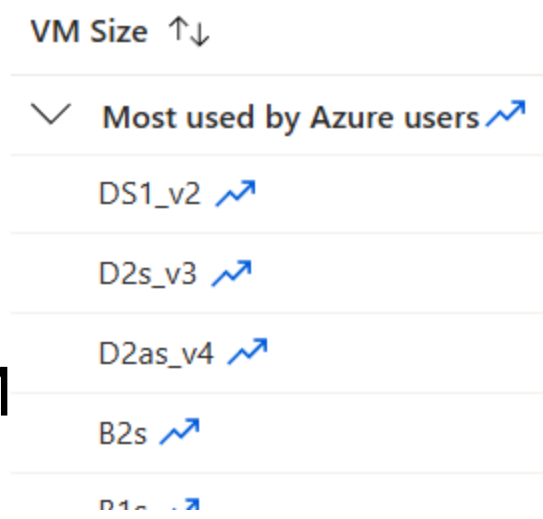
- Azure's "Portal" is the web interface
- portal.azure.com



Azure Terminology

- **Resources** - virtual machines, storage, and SQL databases, etc
- **Resource Group** - logical containers where you can deploy and manage Azure resources
- **Subscription** – Used to manage costs and organize the resources that are created by users, teams, and projects

VM Sizing



- All VM sizes have a number of vCPU and amount of RAM
- Don't worry about the starting disk size, you can always increase the disk size or add a 2nd disk
- Family
  - D – General purpose, the most common type used
  - B – "Burstable", workloads that do not need continuous CPU
  - E – High memory workloads (i.e., high RAM lowish CPU count)
  - F – Higher end CPUs (e.g., AVX512 support)
  - N – GPU VMs
  - L – Ultra-low disk latency, directly mapped to local NVMe storage

Storage vs Databases

- Azure Storage
  - Designed for storing data in the cloud
  - Blob Storage – A general purpose file store, can be used for backup, archiving, serving static assets for web applications, and more
  - File Storage – File Shares, virtual drives, etc. We won't be using this (not directly)
  - Also niche modes such as Tables, Queues, Data Lake Storage
- Azure databases
  - Essentially a managed database, e.g., MongodB, SQL, Maria
  - Databases are typically for storage, retrieval, and updating of small pieces of data at huge transactional volumes

Blob SAS URL

- A way to reference files in Blob storage
- A file can be downloaded directly via HTTP GET or used within an Azure blob storage API (e.g., in Python, JavaScript, C++, Java)

https://myaccountname.blob.core.windows.net/testcontainer/filename.txt?sp=r&st=2024-08-28T04:20:53Z&se=2024-08-28T12:20:53Z&spr=https&sv=2022-11-02&sr=b&sig=1fL4ma9YX1uEaAbRsYZiaXnXnYAi9DpyOc%2BA2jwZqtE%3D

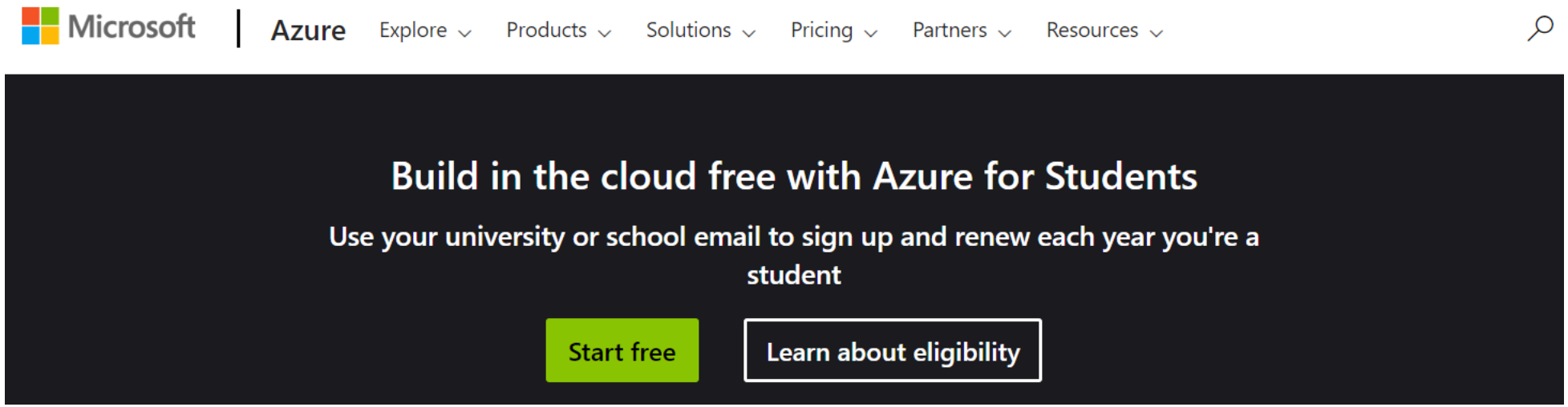
Resource URI https://myaccountname.blob.core.windows.net/testcontainer/filename.txt?

SAS Token
sp=r&
st=2024-08-28T04:20:53Z&
se=2024-08-28T12:20:53Z&
spr=https&
sv=2022-11-02&
sr=b&
sig=1fL4ma9YX1uEaAbRsYZiaXnXnYAi9dpyOc%2BA2jwZqtE%3D

Start of Lab

Creating Student Account

- Google “Azure for Students”, click “Start Free”
- Use your UMD email
- It will have you make an Azure account tied to your UMD email if you don’t already have one



Create a Linux VM

- Subscription – Azure for Students
- Resource Group – Create new
- Region – East US 2
- Image – Ubuntu Server 22.04
- Size – Whatever it lets you choose (e.g., D2s, B1s)
- Authentication type – password
- Fill in the rest to the best of your judgement
- Review + Create, then Create

VM Tweaks

- Go to Resource, or find your VM in the list of resources
- On the left, Networking -> Network Settings
- Create Port Rule button on the right, add an “Inbound port rule”
- Under Service select RDP
- Add
- Repeat above if you don’t already see SSH (22) in the list

Inbound port rules (5)

300	SSH	22	TCP	Any	Any	Allow	
310	AllowAnyRDPInbound	3389	TCP	Any	Any	Allow	

SSH into VM (Screenshot-1)

- On VM's Overview page, copy Public IP address on the right
- In your favorite terminal: ssh azureuser@172.201.151.117
- You'll know it worked if you get the following:

```
azureuser@marc-test-vm:~$
```

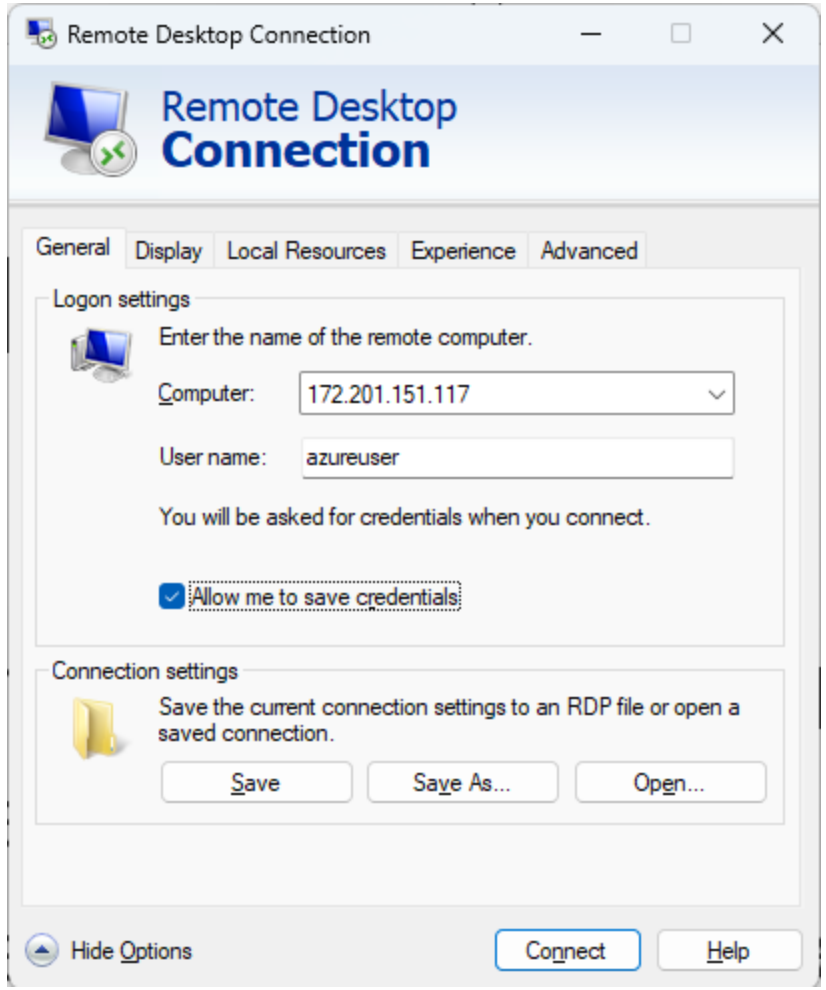
- sudo apt update
- sudo apt install xfce4 xrdp
- <Grab a screenshot>

RDP into VM

- On your laptop open Remote Desktop Connection or similar app
- Check “show options”
- Check “Allow me to save credentials”
- Attempt to RDP into your VM

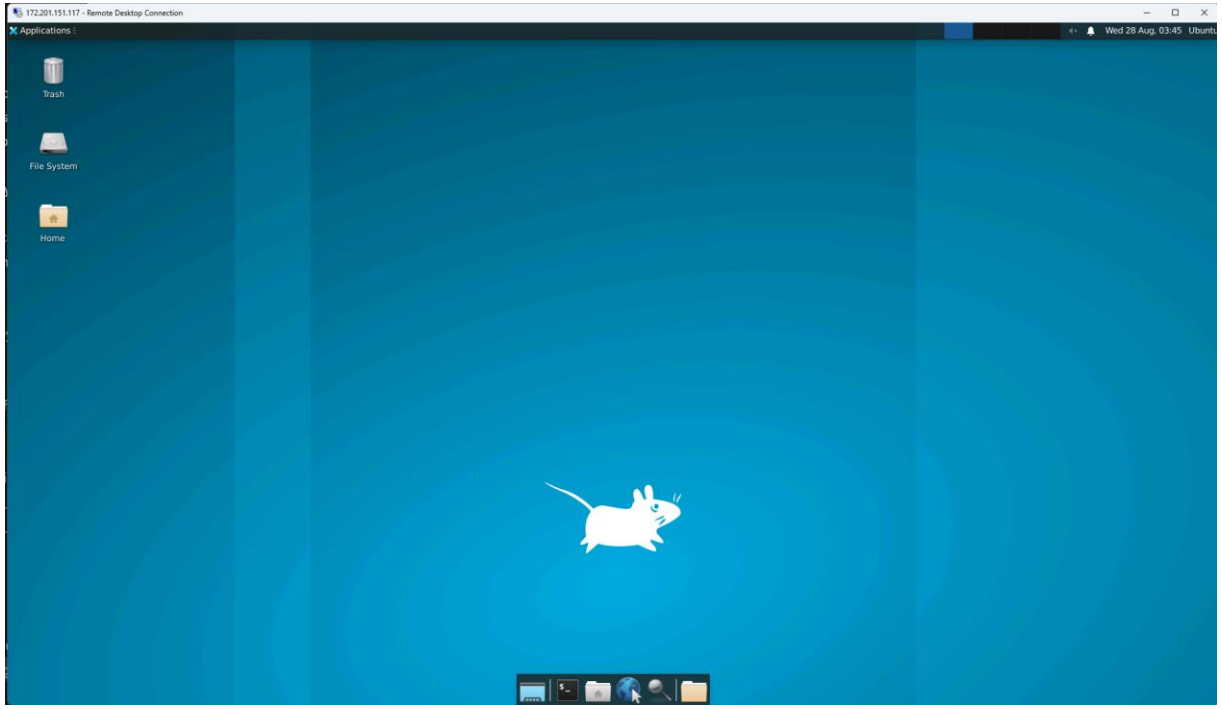
For debugging:

```
azureuser@marc-test-vm:~$ sudo systemctl status xrdp
● xrdp.service - xrdp daemon
     Loaded: loaded (/lib/systemd/system/xrdp.service; enabled; vendor preset: enabled)
     Active: active (running) since Wed 2024-08-28 03:41:33 UTC; 1min 9s ago
       Docs: man:xrdp(8)
             man:xrdp.ini(5)
   Main PID: 6619 (xrdp)
      Tasks: 1 (limit: 1053)
     Memory: 864.0K
        CPU: 14ms
     CGroup: /system.slice/xrdp.service
             └─6619 /usr/sbin/xrdp
```

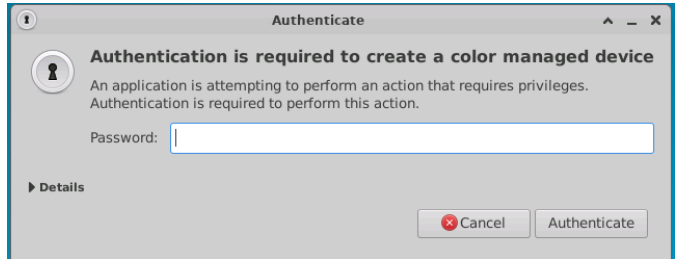


RDP into VM

- Success will look like this ->
- <Grab a screenshot>

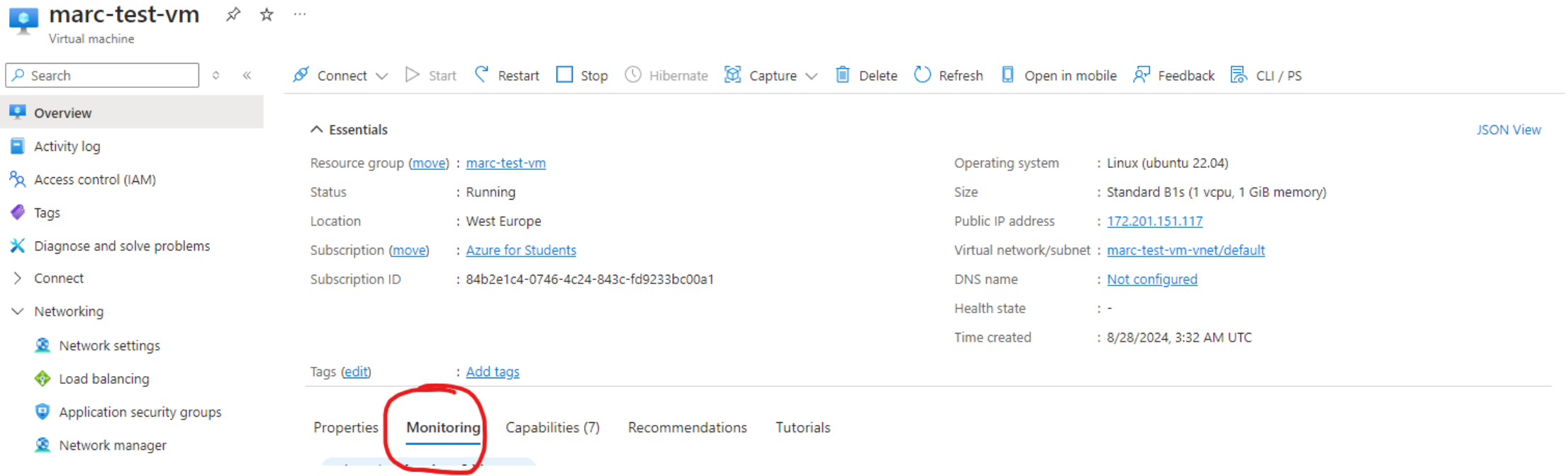


- If you see this window feel free to just close it



Azure Monitors

- Back to Azure Portal, go to your VM resource
- Go to Insights on the left, click Enable, click Configure, continue below
- Within Overview, look for the Monitoring tab towards the middle of the screen
- Click “Show more metrics” expand text
- You should see CPU, Disk, Network



Azure Monitors (Screenshot-2)

- Once Insights has finished deploying, go back to the Insights page and click the Analyze Data button
- You may have to wait another few minutes to be able to see activity
- <Grab a screenshot of the plots>

- Note: <u>Students having problems enabling Azure monitor insights can use the following steps instead:</u>

  - Go back to Azure portal, search Log Analytics workspaces.
  - Create one within the same research group and the same region as your VM.
  - Go back to VM, go to Insights on the left, click Enable.
  - Click 'create new' in data collection rule, in 'Log Analytics workspaces',
    - Select the one created above, click create.
    - Select the created data collection rule, click configure.

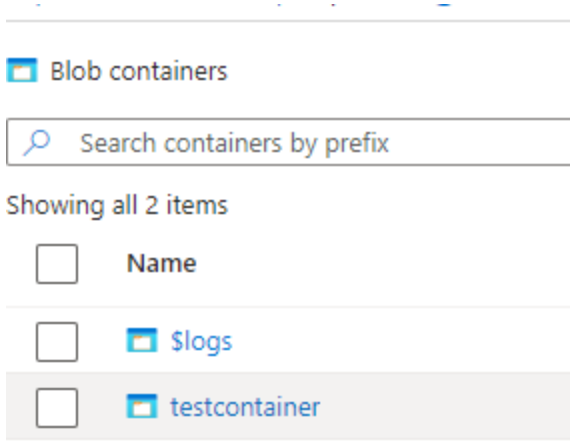
Turn off VM

- When you finished the lab, turn off the VM (shut it down, don’t delete it)
- This will avoid eating into the $100 of credit
- The remaining (non-VM) Azure resources can be left as-is

Storage Account

- Create a Resource, choose Storage Account
- Your storage account name can't include hyphens and must be globally unique across Azure
- Primary Service: Azure Blob Storage
- Primary Workload: Backup and Archive
- Performance: Standard
- Redundancy: LRS
- Click Review and Create, then Create

Uploading a File, then Getting SAS URL (Screenshot -3)



- Go to your Storage Account
- On the left, Storage Browser, Blob Containers
- Add container
- Click the newly created container →
- Upload any file (e.g. create a new .txt file)
- Click the three dots to the right of your file, then click “Generate SAS”
- Click Generate button
- Copy the “Blob SAS URL” and paste it into your browser in a new tab
- Optional: Try downloading the blob without the portion after “?”

Lab Submission

- Submit four screenshots:
  1. Terminal window showing you SSHed into VM
  2. VM Desktop (blue screen with the mouse) (optional)
  3. VM Insights page
  4. Blob storage explorer showing a list of file(s)